

**Problem**

Let  $p$  and  $q$  be two-digit integers, neither divisible by 10. Let  $r$  be the four-digit integer obtained by writing the digits of  $p$  followed by the digits of  $q$ . Thus

$$r = 100p + q.$$

A computer prints  $r$  whenever

$$\gcd(p, q) = 1$$

and

$$p + q \mid r.$$

If the largest printed number is  $N$ , determine the number formed by the last two digits of  $N$ .

*Source: IOQM 2024*

## §1 Extracting the Divisibility Condition

Since

$$r = 100p + q,$$

the condition

$$p + q \mid r$$

becomes

$$p + q \mid 100p + q.$$

Subtracting  $p + q$  from the right-hand side, we obtain

$$p + q \mid 99p.$$

Now

$$\gcd(p, q) = 1.$$

Hence

$$\gcd(p, p + q) = \gcd(p, q) = 1.$$

Since  $p + q$  divides  $99p$  and is coprime to  $p$ , Euclid's lemma implies

$$p + q \mid 99.$$

Therefore

$$p + q$$

must be a positive divisor of 99.

## §2 Bounding the Sum

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The divisors of 99 are

$$1, 3, 9, 11, 33, 99.$$

Because  $p$  and  $q$  are two-digit numbers,

$$p + q \geq 10 + 10 = 20.$$

Thus the only possible value is

$$p + q = 33$$

or

$$p + q = 99.$$

To maximize

$$r = 100p + q,$$

we first maximize  $p$ , since the coefficient of  $p$  is much larger than that of  $q$ .

The largest possible sum is

$$p + q = 99.$$

Hence we seek two-digit integers satisfying

$$p + q = 99, \quad \gcd(p, q) = 1.$$

### §3 Maximizing the Four-Digit Number

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Since

$$q = 99 - p,$$

we have

$$r = 100p + (99 - p) = 99p + 99.$$

Thus maximizing  $r$  is equivalent to maximizing  $p$ .

Because  $q$  must remain a two-digit number not ending in 0,

$$q \geq 10.$$

Therefore

$$p \leq 89.$$

The choice  $p = 89$  gives

$$q = 10,$$

but  $q$  is divisible by 10, which is forbidden.

The next possibility is

$$p = 88, \quad q = 11,$$

yet

$$\gcd(88, 11) = 11 \neq 1.$$

Next,

$$p = 87, \quad q = 12,$$

and

$$\gcd(87, 12) = 3 \neq 1.$$

Finally,

$$p = 86, \quad q = 13,$$

for which

$$\gcd(86, 13) = 1.$$

Hence

$$r = 100 \cdot 86 + 13 = 8613.$$

Since we have chosen the largest admissible value of  $p$ , this is the largest possible printed number.

## §4 The Final Conclusion

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Thus

$$N = 8613.$$

The number formed by the last two digits of  $N$  is

$$\boxed{13}.$$